FI FISHINTEL GLOBAL

ENVIRONMENTAL RECONSTRUCTION REPORT

Decision-Time Environmental Reconstruction

Capsize of Sailing Yacht During Convective Storm Event

INCIDENT WINDOW	18–19 August 2024
REPORT REFERENCE	FIG-FLAGSHIP-003
DECISION FRAME	02:00 UTC / 04:00 Local
REPORT VERSION	v3.0 — Chain v1.0
VESSEL CATEGORY	Private sailing yacht, 55.9 m
CLASSIFICATION	Public — Anonymised
OPERATIONAL CONTEXT	At anchor — 22 POB

HAZARD

Capsize / stability failure

PIPELINE VERIFIED — EVIDENCE CHAIN VERIFIED — 27 ARTIFACTS SEALED

PUBLIC VERSION — ANONYMISED CASE STUDY — VESSEL AND INDIVIDUALS NOT IDENTIFIED

01

Executive Summary

Incident type: Capsize following sudden convective wind event. A 56-metre privately operated sailing yacht, anchored overnight off the northern coast of Sicily, capsized and sank during a mesoscale convective downdraft. Seven of 22 persons on board lost their lives.

Core question: What environmental conditions were objectively knowable at the time the vessel was at anchor, and what was the operational margin before the situation became irreversible?

Method: ERA5 reanalysis data was extracted at the incident coordinates across a five-point time window (23:00–03:00 UTC) and processed through the FishIntel deterministic reconstruction pipeline with FIG_UNIT_GATE validated conversion and T_irr decision-window analysis.

Key result: ERA5 shows benign synoptic conditions: 5.53–6.19 kt wind, 0.368–0.410 m seas. The reconstruction runner classifies all timestamps as LOW risk. However, the T_irr framework classifies the operational situation as `REGIME 2 — REGIME_2_DANGER` with a conservative margin of only 1.2 min. The capsize was caused by a mesoscale downdraft exceeding 70 kt that ERA5 cannot resolve.

Analytical finding: The margin between operational viability and irreversible loss was already critically thin at the ERA5 synoptic baseline. The mesoscale downdraft did not gradually erode this margin — it

exceeded it instantaneously. The vessel heeled to 90° in under 15 seconds.

DECISION SNAPSHOT

LOCATION	DATE	PERSONS ON BOARD
Northern coast of Sicily	19 August 2024	22
FATALITIES	**ERA5 BASELINE WIND**	**ERA5 WAVE HEIGHT**
7	6.28 kt	0.392 m
OBSERVED EVENT	**CONSERVATIVE MARGIN**	**RISK REGIME**
Convective downdraft >70 kt	1.2 min	REGIME 2 — REGIME_2_DANGER

02

Incident Overview

Vessel Profile

TYPE	Private sailing yacht
LOA	55.9 m
CONSTRUCTION	Aluminium alloy
MAST HEIGHT	72 m
PERSONS ON BOARD	22 (guests and crew)
CONFIGURATION	Motoring — centreboard raised, no sails set
ENGINE STATUS	OPERATIONAL — generators, steering and hydraulic pumps running
HAZARD CLASSIFICATION	CAPSIZE_STABILITY (FIG Step 11, exact match)

On the evening of 18 August 2024, the vessel anchored approximately 0.5 nm south-east of a harbour on the northern coast of Sicily. According to the MAIB interim report (p.2), weather forecasts available to the crew indicated the possibility of thunderstorm activity in the region.

From approximately 01:00 local time, crew members observed occasional flashes of lightning to the west. By 03:00, the second deckhand noted that thunderclouds and lightning appeared to be getting closer. At 03:55, the advancing storm front was recorded on video (MAIB interim report, p.3).

The chief engineer started all three generators, then went to the wheelhouse to start steering pumps and hydraulic pumps for the controllable pitch propeller. The chief officer confirmed that the vessel was ready to manoeuvre (MAIB interim report, p.3–4).

At approximately 04:06 local time (02:06 UTC), a sudden wind increase exceeding 70 knots caused the vessel to heel to 90° starboard in under 15 seconds. The vessel subsequently sank in approximately 50 metres of water.

03

Decision-Time Environment

The following chronology presents the sequence of events from the crew's first observation of the approaching storm through the capsize, with environmental context at each stage. Markers indicate decision points and critical events.

18 Aug — 21:24 local (19:24 UTC)

Vessel anchored in the bay

Anchored east of the main breakwater, approximately 0.5 nm south-east of harbour. 22 persons on board. Motoring configuration.

```
ERA5 23:00 UTC: 5.53 kt sustained, 12.39 kt gusts, 0.368 m SWH
```

19 Aug — 01:00 local (23:00 UTC)

Lightning observed to the west

Crew observed occasional flashes of lightning away to the west (MAIB interim report, p.2). Synoptic conditions remain benign.

```
ERA5 23:00 UTC: 5.53 kt sustained | Decision window: OPEN
```

19 Aug — 03:00 local (01:00 UTC)

Thunderclouds approaching

Second deckhand noted that thunderclouds and lightning appeared to be getting closer (MAIB interim report, p.3). Still no significant wind change at synoptic scale.

```
ERA5 01:00 UTC: 5.54 kt sustained, 11.46 kt gusts | Decision
window: OPEN
```

19 Aug — 03:55 local (01:55 UTC)

Storm front recorded on video

Second deckhand videoed the advancing storm from the wheelhouse (MAIB interim report, p.3). Chief engineer started all three generators. Steering pumps and hydraulic pumps started.

```
Vessel preparing to manoeuvre | Decision window: NARROWING
```

19 Aug — ~04:00 local (02:00 UTC)

Decision frame — vessel ready to manoeuvre

Chief officer confirmed the vessel was ready to manoeuvre. All generators running. This is the ERA5 decision-frame timestamp.

```
ERA5 02:00 UTC: 6.28 kt sustained, 12.60 kt gusts, 0.392 m SWH |
Decision window: NARROWING
```

19 Aug — 04:06 local (02:06 UTC)

Extreme downdraft — capsize

Sudden wind increase exceeding 70 knots. Vessel heeled to 90° starboard in under 15 seconds. Subsequently sank in approximately 50 metres of water. Seven fatalities.

```
Mesoscale convective downdraft >70 kt | NOT RESOLVED by ERA5 |
Decision window: CLOSED
```

04

Environmental Reconstruction

> **ERA5 Resolution Constraint.** ERA5 at 31 km horizontal and 1-hour temporal resolution captures the synoptic background only. It cannot resolve the mesoscale convective downdraft that caused the capsize. Values below represent the pre-storm baseline, not peak event conditions.

UTC	LOCAL	WIND (KT)	GUST (KT)	WAVE (M)	PRESSURE	DIR
23:00	01:00	5.53	12.39	0.368	1009.5	298
00:00	02:00	4.94	10.37	0.372	1009.1	279
01:00	03:00	5.54	11.46	0.381	1008.2	264
02:00	**04:00** ◂ DECISION FRAME	**6.28**	**12.60**	**0.392**	**1008.1**	**263**
03:00	05:00	6.19	13.05	0.410	1008.1	268

6.28 kt	12.60 kt	0.392 m	1008 hPa
WIND AT DECISION FRAME	GUST AT DECISION FRAME	WAVE HEIGHT	PRESSURE (MSL)

INTERPRETATION

The synoptic environment was characterised by light westerly winds, calm seas, and a gentle pressure trough — conditions consistent with a pre-convective Mediterranean summer night. The environmental baseline gave no indication of the catastrophic wind event that followed.

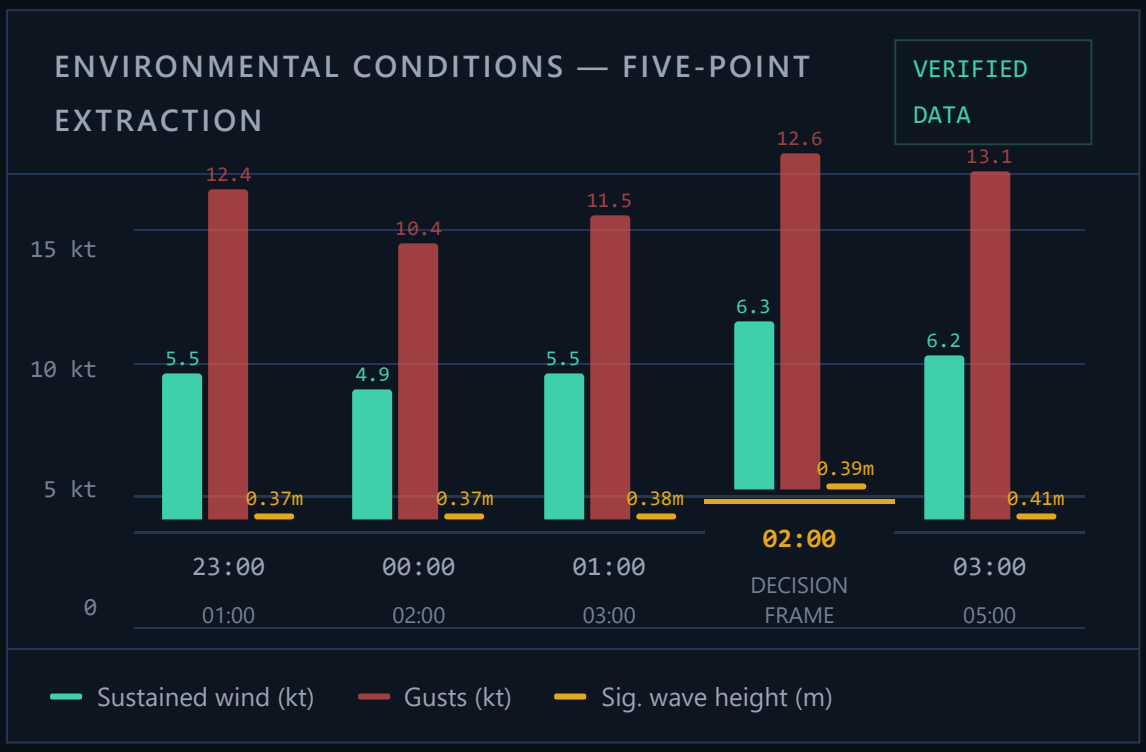


CHART NOTE

All five extraction points show benign synoptic conditions — sustained winds below 7 knots, gusts below 14 knots, and seas below half a metre. The scale (0–15 kt) reflects the actual ERA5 range. The mesoscale downdraft that caused the capsize (>70 kt) is entirely outside this range and is not resolved by ERA5.

05

Vessel Stability Context

ASSUMPTION-BASED — DESKTOP ASSESSMENT ONLY

All stability values below are from the MAIB interim report (May 2025), referencing a desktop assessment by the Wolfson Unit. Assumes motoring configuration, centreboard raised, 10% consumables, symmetrical wind loading. **Not confirmed by physical examination.** No values enter the evidence chain.

The following values are quoted from the MAIB interim report's desktop stability assessment and are not generated by the FishIntel evidence chain.

PARAMETER	VALUE	STATUS	MAIB SOURCE
Angle of vanishing stability	70.6°	ASSUMPTION	p.7 — Wolfson Unit
Limiting steady wind (high)	44.8 kt	ASSUMPTION	Table 1
Limiting gust wind (high)	63.4 kt	ASSUMPTION	Table 1
Mast heeling contribution	~50%	ASSUMPTION	p.7

Vulnerable wind angle	60°–100° off bow	ASSUMPTION	p.7
Time to 90° heel	<15 seconds	Witness	p.4

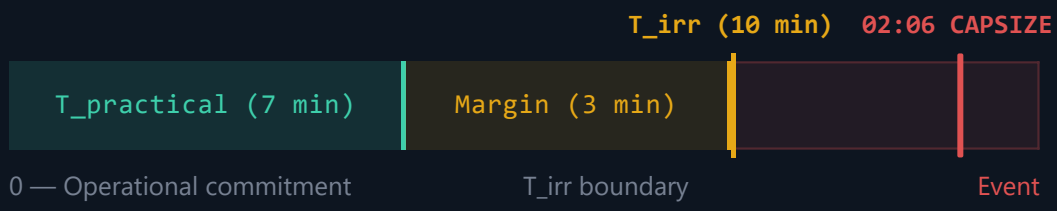
06

Decision Window Assessment

FishIntel's Time-to-Irreversibility (T_{irr}) framework identifies when an operational situation becomes irreversible — the point beyond which no reasonable intervention can change the outcome.

Decision Window — Reversibility Timeline

Time from operational commitment to irreversibility at ERA5 baseline conditions.
Decision frame: 02:00 UTC (04:00 local) — 6 minutes before capsize.



7 min	10 min	3 min	1 min
T_PRACTICAL	T_IRR EFFECTIVE	MARGIN (BEST)	MARGIN (CONSERVATIVE)

Decision Window Findings

RISK REGIME	REGIME 2 — REGIME_2_DANGER
MECHANISM	GRADUAL_DEGRADATION

T_IRR (EFFECTIVE)	10.3 min (617,333 ms)
T_IRR (CONSERVATIVE)	8.2 min (493,866 ms)
T_PRACTICAL	7.0 min (420,000 ms)
MARGIN (BEST)	3.3 min (197,333 ms)
MARGIN (CONSERVATIVE)	1.2 min (73,866 ms)
CONVERGENT SQUEEZE	Not detected
DECISION GATES	3
DECISION FRAME	02:00 UTC / 04:00 local — 6 minutes before capsize

INTERPRETATION

Even at the benign ERA5 synoptic baseline, the T_irr framework classifies this as Regime 2 — Danger. The conservative margin of 1.2 min represents the minimum time between operational recognition and irreversible loss. When the mesoscale downdraft arrived (reported >70 knots), this margin was not gradually eroded — it was instantaneously exceeded. The vessel heeled to 90° in under 15 seconds, a timescale that renders any intervention physically impossible.

07

Risk Analysis

TIMESTAMP	RISK SCORE	LEVEL	DECISION
2024-08-18T23:00:00	32.1	LOW	ACCEPTABLE
2024-08-19T00:00:00	25.9	LOW	ACCEPTABLE
2024-08-19T01:00:00	26.9	LOW	ACCEPTABLE
2024-08-19T02:00:00	28.0	LOW	ACCEPTABLE
2024-08-19T03:00:00	31.0	LOW	ACCEPTABLE

ERA5 SYNOPTIC BASELINE

6.28 kt

Sustained wind at decision frame

LOW

All 5 timestamps — acceptable

ACTUAL EVENT (OUTSIDE ERA5)

>70 kt

Mesoscale convective downdraft

CATASTROPHIC

Capsize in <15 sec — 7 fatalities

08

Key Analytical Finding

PRIMARY FINDING

The ERA5 synoptic environment was benign. The catastrophe was driven by a mesoscale event embedded within it.

At 02:00 UTC, ERA5 reports 6.28 kt sustained wind and 0.392 m seas. By every synoptic metric, conditions appeared acceptable. Yet the T_irr framework classifies the operational situation as Regime 2 — REGIME_2_DANGER with a conservative margin of only 1.2 min.

The operational configuration — an anchored vessel at night, 22 persons aboard, a visible approaching storm, and a stability profile (assumption-based) assessed as vulnerable to gusts exceeding 63.4 kt — created a situation where the margin between viability and catastrophe was already critically thin at the synoptic baseline.

When the mesoscale convective downdraft arrived (reported exceeding 70 knots), the conservative margin of approximately 1.2 minutes was not gradually eroded. It was instantaneously exceeded by an order of magnitude. The vessel heeled to 90° in under 15 seconds.

The foreseeability question is not whether the synoptic conditions were dangerous — they were not. The question is whether the *potential for extreme convective winds* was identifiable from the synoptic pattern and available forecasts. The Met Office post-incident study, referenced in the MAIB interim report (p.8), concluded that a mesocyclonic storm was highly likely in the prevailing conditions, with downdrafts potentially reaching well in excess of 64 knots.

This report does not attribute fault or negligence to any individual. Its purpose is to quantify the environmental context in which operational decisions were made.

09

Evidence Integrity

This report is a rendering of outputs produced by the FishIntel deterministic reconstruction pipeline. All environmental values are traceable through an auditable cryptographic chain. No manual adjustments or subjective corrections are applied.

HASH CHAIN — V1.0

SOURCE INTEGRITY

ERA5 ATMO SEAL	e6eb7508e4c4ac1acd84742f60f9c252861969244cea9a512637c63de73a7103
ERA5 WAVE SEAL	7b2051d2855f62775c05288e709e3cbec94bfa7b3b2a4107d8eecb09d001f0f8
UNIT GATE CONFIG	989aadf7b9b1a4acef8606e162015337e75a8ce5fcb4edfb1e0d79f4d59513aa

MODEL INTEGRITY

CORE V7 INPUT	2d98a77220fde67abe42fc9a99671e9fddea417ee51c8490d38ae6147ba9407c
CORE V7 AUDIT	46c2761573cbadda4ab84ae59b6df62b625ae539ddb744ccf911bebfbb49c7ab
T_IRR HASH	c88a86163221e9c274e3f4d3a6da59cd32a10360735bc6ee7f3b22d158337954

CHAIN INTEGRITY

COMBINED HASH	547df3c003733d60f6bd1e0763d3b274a8e7bebcd96937b0afc40237c7c28981
PROTOCOL HASH	a3e76b4ad166c65ec870564e9774acf99469088d82be8e61e5d6f8c878eab230
AUDIT PACK HASH	79106dd2f3b9c8c112582f05658f28153e42a6db4fbf1f577cd848fdc1f809ef

Methodology

FishIntel Global reconstructs environmental conditions using ECMWF ERA5 atmospheric reanalysis — the same dataset relied upon by meteorological agencies, climate researchers, and aviation safety authorities worldwide. ERA5 provides hourly estimates on a global grid at approximately 31 km horizontal resolution.

All values pass through a unit-gated validation layer enforcing explicit conversion factors and plausibility bounds using Decimal arithmetic. The pipeline is deterministic: identical inputs always produce identical outputs.

This reconstruction operates within defined boundaries. ERA5 does not capture localised effects smaller than its grid resolution. Where a case involves mesoscale phenomena, the reconstruction presents ERA5 as the synoptic baseline, not the peak event conditions.

FishIntel Analytical Architecture

FishIntel operates as a multi-system analytical architecture designed to reconstruct the environmental conditions surrounding maritime incidents and quantify the operational margin before irreversible outcomes occur.

The architecture consists of several interlocking analytical systems:

ARCHITECTURE COMPONENTS

Source Extraction System — converts investigation material and case information into structured machine-readable inputs.

Environmental Reconstruction System — retrieves and reconstructs decision-time environmental conditions using ERA5 atmospheric reanalysis.

Unit Integrity System — validates and canonicalises all physical measurements using explicit conversion factors and Decimal arithmetic.

Risk Inference System — evaluates reconstructed environmental conditions to produce structured risk assessments.

Time-to-Irreversibility (T_irr) System — calculates the operational window before intervention becomes ineffective.

Evidence Chain System — generates deterministic seals, hashes, and verification artifacts for every analytical step.

Report Generation System — renders validated outputs into investigation-grade analytical reports.

These systems operate together within a deterministic reconstruction pipeline.

Audited Implementation Modules

The modules listed below are audited implementation components supporting the FishIntel analytical architecture.

MODULE	VERSION	HASH

FIG_CANONICAL.py	1.1.1	b9122b47becd2c4253201f0c8c76…
FIG_HAZARD_CATEGORY.py	1.0.0	8b80a12062978542af0bcd1f38d8…
FIG_CASE_CONTEXT.py	1.0.0	b388d8adeb0a9844c3670454ac39…
FIG_T_IRR_V1.py	1.0.0	4e8648c48aece2b4d1894dfed3a4…
FIG_UNIT_GATE.py	1.0.0	30c61d2e3e63116eb5f7aacbd341…
FIG_SOURCE_SEAL.py	1.0.0	f1be9157b7f1133a397a3199a48e…
FIG_COMBINED_EVIDENCE.py	2.0.0	b9d2407477d721467bbcbc356b6b…
FIG_AUDIT_PACK.py	1.0.0	06f8eeeca55a550e6f824dbec11d…
FIG_CORE_V7_WEAPON.py	7.4	6b2dc01add11e29b1612b90f86a1…

11

Limitations and Caveats

ERA5 RESOLUTION

31 km horizontal, 1-hour temporal. Cannot resolve mesoscale convective downdraft. Synoptic background only.

MAIB INTERIM REPORT

Interim publication (May 2025). Criminal investigation restricts access. Findings may change.

STABILITY ASSESSMENT

Desktop assumptions only. Unverified. Not in evidence chain.

PROVISIONAL COORDINATES

38.076°N, 13.538°E estimated. Immaterial at ERA5 31 km grid.

NO CROSS-VALIDATION

ERA5 only. v1.0 chain (9 components). No independent GFS verification.

VESSEL TYPE

FIG vocabulary v1.0 has no yacht category. Classified as OTHER.

Closing Statement

Maritime incidents involving mesoscale weather events present a particular challenge for environmental reconstruction. The synoptic environment may appear benign by every conventional metric, yet the conditions for extreme localised wind events may already be present — identifiable from forecast products but invisible to coarse-resolution reanalysis data.

Environmental reconstruction cannot resolve these mesoscale events directly. What it can do is establish the synoptic baseline with precision, quantify the operational margin that existed before the event arrived, and assess whether the potential for such an event was foreseeable from the data and forecasts available at the time.

In this case, the synoptic conditions were benign. The stability margin was critically thin. The mesoscale event that arrived was of a type and intensity that the Met Office post-incident study, referenced in the MAIB interim report, assessed as highly likely in the prevailing conditions. The conservative intervention margin of 1.2 min was exceeded instantaneously. The question this reconstruction leaves for investigators, insurers, and regulators is whether the conditions that made this outcome possible were objectively knowable before it occurred.

Appendix

Evidence Manifest

27 artifacts. Every numeric value programmatically extracted from these files.

FILENAME	SIZE	DIRECTORY
audit_pack.json	3,567	evidence_flagship003/
case_context.json	2,884	evidence_flagship003/
combined_evidence.json	2,261	evidence_flagship003/
combined_legal_evidence.txt	2,942	evidence_flagship003/
era5_atmo_seal.json	865	evidence_flagship003/
era5_wave_seal.json	797	evidence_flagship003/
gated_values_0000.json	904	evidence_flagship003/
gated_values_0100.json	904	evidence_flagship003/
gated_values_0200.json	904	evidence_flagship003/
gated_values_0300.json	904	evidence_flagship003/
gated_values_2300.json	904	evidence_flagship003/
hazard_classification.json	241	evidence_flagship003/

prediction_0000.json	4,000	evidence_flagship003/
prediction_0100.json	3,996	evidence_flagship003/
prediction_0200.json	3,994	evidence_flagship003/
prediction_0300.json	3,994	evidence_flagship003/
prediction_2300.json	4,000	evidence_flagship003/
raw_era5_0000.json	2,257	evidence_flagship003/
raw_era5_0100.json	2,255	evidence_flagship003/
raw_era5_0200.json	2,254	evidence_flagship003/
raw_era5_0300.json	2,255	evidence_flagship003/
raw_era5_2300.json	2,259	evidence_flagship003/
tirr_analysis.json	803	evidence_flagship003/
tirr_legal_evidence.txt	2,603	evidence_flagship003/
reconstruction_results.csv	666	reconstruction_results_flagship003/
reconstruction_results.json	3,937	reconstruction_results_flagship003/
reconstruction_summary.txt	338	reconstruction_results_flagship003/

FishIntel Global — Maritime Foreseeability Intelligence

This report is generated from verified FishIntel reconstruction pipeline outputs.

Environmental reconstruction methodology & system: © FishIntel Global (FIG LTD).

Source investigation data: Crown copyright (MAIB). Used in accordance with published reproduction terms.

Combined hash: `547df3c003733d60f6bd1e0763d3b274a8e7bebcd96937b0afc40237c7c28981` — 2026-04-03T01:36:56Z